

Measure-Valued Stochastic Control and Matching in Spatial Forestry Markets

EA Forests Research

October 2026

Abstract

We formulate a partially observed stochastic control problem for a spatial forestry market. The physical state is a finite measure over tree attributes, locations, log recoveries, processor specifications, and operating capacities. Conditional McKean–Vlasov dynamics describe growth and quality under common and idiosyncratic noise. Harvest and bucking induce a finite log measure. A capacitated partial-transport problem allocates heterogeneous logs to processors through spatial operators, and its dual variables measure the marginal value of compatible supply, demand, and capacity. The decision state is the joint posterior over physical state and structural parameters. We give the associated belief-state recursion, a Wasserstein HJB subproblem, a risk-sensitive extension, and a finite-dimensional commercial projection. An East African prior-predictive calibration illustrates the projection and records the remaining identification requirements.

Keywords: stochastic control; McKean–Vlasov dynamics; Wasserstein geometry; unbalanced optimal transport; Bayesian experimental design; forestry markets.

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1 The state of a spatial forestry market

1. The central modelling problem

A forestry market couples biological growth, harvesting, processing constraints, transport, and information. Biological state determines feasible material. Harvesting maps trees into logs. Processor specifications assign value to regions of log space. Spatial operators determine delivered cost. Observations update the decision maker’s information.

A useful mathematical model must therefore preserve heterogeneity long enough for the economically important compatibility tests to be applied.

Suppose two estates each contain 100 hectares and 20,000 cubic metres of standing volume. Their commercial values can differ sharply. Estate A could contain mostly logs above a target processor’s small-end diameter threshold and lie 20 km from a mill. Estate B could contain the same total volume in smaller stems, more defect, and a 250 km transport route. The state variables that explain the difference have been averaged away by the scalar summary.

The paper’s response is to make the primary state **measure-valued**. A measure records how much mass or how many objects occupy each region of a state space. For forestry, that state space can carry location, age, log-diameter and height, species or genetics, quality, and management condition.

The standing-tree object is written schematically as

$$\mu_t \in \mathcal{M}_+(E_T),$$

where E_T is the marked tree space and $\mathcal{M}_+(E_T)$ is the cone of finite nonnegative Radon measures on that space.

The total mass $\mu_t(E_T)$ represents an intensity, count, or area-weighted count, according to the sampling convention. Normalization would discard that quantity.

2. Why there are three nested problems

2.1 State estimation

Plots, remote-sensing features, harvest records, processor intake, and contractor performance provide partial observations of the latent economic state.

The model therefore includes an observation process and a posterior state. A map pixel supplies evidence about a latent tree distribution; a calibrated observation model is required to infer DBH, log grade, or harvest value.

2.2 Matching

Once harvestable material has been represented as a log distribution, economic usefulness is determined by compatibility. Processor j may accept only a subset S_j of log space. Operator k may have finite capacity or equipment constraints. Delivered cost depends on the log, route, and operator.

An average-price identity such as

$$\text{revenue} = \text{tonnes} \times \text{average price}.$$

omits the relevant constraints. The allocation problem acts on feasible triples

$$(\lambda, j, k),$$

where λ is a log state, j a processor, and k an operator.

2.3 Sequential control

A decision maker can change both the current physical/economic outcome and the quality of future information. Choosing an inventory campaign, instrumenting a contractor, testing a processor's grade response, or exposing a new supply cluster to the market can reveal parameters that affect later decisions.

This makes the problem a dual-control problem: actions can have both immediate payoff and learning value.

3. The operator chain

The paper is easier to understand when written as a sequence of operators.

3.1 Biological evolution

A stochastic evolution operator moves the standing-tree measure through size and quality space and changes its mass through mortality and recruitment.

3.2 Harvest and bucking

A harvest rule removes mass from the standing-tree measure. A bucking kernel maps each harvested tree state to a finite measure of logs.

Schematically,

$$\mu_t \longrightarrow \Lambda_t,$$

where Λ_t is a log measure.

3.3 Compatibility and allocation

Processor specifications and operating constraints define which parts of Λ_t are commercially useful. A partial-transport problem chooses an allocation measure Γ_t .

$$\Lambda_t \longrightarrow \Gamma_t.$$

The allocation may leave logs unmatched, demand unmet, and operator capacity idle. These inequalities define partial matching.

3.4 Observation and learning

Observations update a posterior over physical state and structural parameters. That posterior becomes the sufficient state for sequential decisions.

The full conceptual loop is therefore

```
latent tree state
-> growth / mortality / recruitment
-> harvest / bucking
-> log distribution
-> processor/operator matching
```

```
-> economic outcomes and observations
-> posterior update
-> next decision
```

4. Distinct economic objects

Several distinctions are essential.

Biological productivity and commercial availability

Fast growth can produce logs outside the target specifications.

Commercial availability and processor compatibility

A large harvestable log volume may be incompatible with a processor's diameter, length, species, moisture, or defect constraints.

Compatibility and delivered surplus

A compatible log can still be uneconomic if transport and operating costs erase its value.

Delivered surplus and negotiated forest-gate price

The seller's outside option and bargaining environment can affect the realized price even after technical matching is known.

Physical coverage and information value

The value of mapped hectares depends on their effect on a decision. Useful observations reduce uncertainty in decision-relevant regions of state space.

5. Structural model versus reduced-order model

The structural model is intentionally richer than current data can fully identify. The paper later introduces a reduced-order commercial projection based on hectares, paying accounts, matched supply, and operating cost.

The models are related by

$$\text{structural state} \xrightarrow{\text{projection}} \text{reduced model},$$

The reduced model is an operational approximation. A scalar “visible tonnes” parameter can be used until data support estimated growth, bucking, and matching operators.

6. Sequential control

The decision maker acts while estimates remain uncertain. Sensing, contracting, and market exposure affect current payoffs and future information.

The action space can include:

- sensing or field verification;
- designed experiments;
- matching exposure;

- contract or posted-price variables;
- coverage intensity; and
- discrete regime changes.

Thus the state must be defined with the eventual decision problem in mind. There is little value in estimating a variable accurately if no feasible action changes as a result. Conversely, a noisy variable near a processor grade threshold can be highly decision-relevant.

7. State, observations, and implementation

The model separates **state definition**, **observation**, and **implementation**. Individual-tree IDs support repeated measurements and remain outside the limiting measure. Earth-observation features enter through the observation model and become tree attributes only after calibration.

Scalar and discrete examples are explicit projections of the structural model.

8. Mathematical scope

The model preserves the distribution of economically relevant states through the biological, technical, and market operators. Measure-valued dynamics describe the physical state, unbalanced transport describes feasible allocation, and Bayesian control describes sequential decisions under partial observation.

2 Geometry of the forestry state

1. The marked tree space

Let (D, d_D) be a compact geographic metric space; diameter is denoted by d . A model instance may take D to be a region, country, estate, or bounded study landscape. A particular forest is represented by where the standing-tree measure has support inside D .

Let:

- G be a finite species/genetic-class set;
- Q_T be a compact tree-quality space;
- Q_L be a compact log-quality space;
- M be a compact management-state space.

To preserve positivity of diameter and height, the paper writes size in log-coordinates $s = (s_d, s_h) = (\log d, \log h)$. A tree state is

$$\xi = (\ell, a, s, g, q, m) \in E_T = D \times \mathbb{R}_+ \times \mathbb{R}^2 \times G \times Q_T \times M.$$

Age remains an explicit coordinate. Two trees can have similar diameter and height but different age, growth potential, mortality risk, or management timing. If a future reduced model shows that age can safely be omitted, that is an empirical model reduction.

2. The log, processor, and operator spaces

A harvested log is represented as

$$\lambda = (\ell, d_s, d_l, L_L, g, q_L, w) \in E_L,$$

where d_s and d_l are small- and large-end diameters, L_L is length, q_L is quality, and w is moisture or another density-relevant state.

Log mass is

$$m_L(\lambda) = \varrho(g, q_L, w) V_L(d_s, d_l, L_L, q_L).$$

Density and recoverable volume depend on state, so the model uses the mass map above in place of a universal cubic-metre-to-tonne constant.

A processor state records location, normalized desired procurement flow, specification parameters, utilization, and a sourcing-radius parameter. An operator state records location, capacity, productivity, equipment/fuel intensity, cost, and compliance/quality information.

These state spaces are assumed Polish. That gives the measurable and topological structure needed for probability measures, weak convergence, selection theorems, and dynamic programming.

3. Why discrete biological marks are stratified

Species and genetic class are discrete marks. If $g=1$ and $g=2$ are species labels, interpolation through $g=1.5$ has no biological meaning.

The paper therefore identifies the tree space with a finite disjoint union of continuous strata:

$$E_T = \bigsqcup_{g \in G} E_T^g.$$

A standing-tree measure can be written as a vector

$$\mu = (\mu^g)_{g \in G}.$$

Continuous growth occurs within a biological stratum. A species change or reclassification, if scientifically meaningful, must be represented by an explicit discrete kernel. A continuous species axis has no biological meaning.

4. Finite measure versus normalized probability law

The persistent tree state is

$$\mu_t \in \mathcal{M}_+(E_T).$$

Its total mass $\mu_t(E_T)$ matters. Harvest and mortality remove mass; recruitment adds mass. When total mass is positive, define the normalized law

$$\bar{\mu}_t = \frac{\mu_t}{\mu_t(E_T)}.$$

These two objects answer different questions:

- μ_t says both **how much** tree mass/count is present and **how it is distributed**;
- $\bar{\mu}_t$ says **what the composition looks like conditional on being in the population**.

Normalizing too early destroys information. A plantation with 100 surviving trees and one with 1,000 surviving trees can have the same normalized size composition.

5. Wasserstein geometry

For probability measures with finite second moment, the quadratic Wasserstein distance is

$$W_2^2(\mu, \nu) = \inf_{\gamma \in \Pi(\mu, \nu)} \int d(x, y)^2 \gamma(dx, dy).$$

It asks for the least quadratic cost of transporting one probability law into another. This makes it natural for growth in continuous state coordinates: mass moves from smaller to larger sizes, or through quality space, while the total probability remains one.

The formal tangent geometry gives the continuity equation

$$\partial_t \mu_t + \nabla \cdot (v_t \mu_t) = 0.$$

That equation conserves total mass. The full forestry state also requires mass change.

6. Hellinger-Kantorovich / unbalanced transport geometry

A mass-changing continuity equation has the form

$$\partial_t \mu_t + \nabla \cdot (v_t \mu_t) = \alpha_t \mu_t + B_t,$$

where α_t is a signed reaction rate and B_t is an external source measure. Mortality or harvest corresponds to negative reaction; planting or recruitment is a source.

Hellinger-Kantorovich geometry is designed for exactly this combination of transport and reaction. It can compare finite measures with different total mass.

The entropy-transport and dynamic formulations used here follow [Liero et al. \[2018\]](#) and [Chizat et al. \[2018\]](#). Entropic approximations can be computed by the scaling algorithms of [Chizat et al. \[2016\]](#).

The conceptual split is therefore:

Wasserstein:	composition of a normalized surviving population
HK/unbalanced:	physical state when mass can appear or disappear

Each geometry therefore serves a separate state representation.

7. A useful normalized-selection identity

Diffusion and selection produce different terms. Consider a simplified finite measure satisfying

$$\frac{d}{dt} \langle \phi, \mu_t \rangle = \langle L\phi, \mu_t \rangle - \langle \lambda \phi, \mu_t \rangle,$$

with state-dependent removal hazard λ . Let $M_t = \mu_t(E)$ and $\bar{\mu}_t = \mu_t / M_t$. Differentiating the normalized expectation gives

$$\frac{d}{dt} \langle \phi, \bar{\mu}_t \rangle = \langle L\phi, \bar{\mu}_t \rangle - \text{Cov}_{\bar{\mu}_t}(\phi, \lambda).$$

The covariance term is a selection effect: if large trees are harvested at a higher rate, the composition of survivors changes even without any biological shrinkage. Recruitment adds further composition terms.

The representative-survivor law determines composition; the finite measure also determines population mass.

8. Linear convexity versus geodesic convexity

$M_+(E)$ is a convex cone under ordinary mixtures

$$(1 - \theta)\mu_0 + \theta\mu_1.$$

This is **linear convexity**. It is useful for resource measures and relaxed controls.

Wasserstein or HK spaces may also have geodesics. A functional can be convex along those geodesics, which is a different statement. Geodesic convexity is relevant to transport and gradient-flow methods; linear convexity is relevant to ordinary convex programs.

Hard grade thresholds, species labels, regime switches, and impulses preclude a global geodesic-convexity assumption.

9. Finite particle approximation and tree identity

A finite interacting system can be written

$$X_t^{i,N}, \quad i = 1, \dots, N,$$

with empirical law

$$\mu_t^N = \frac{1}{N} \sum_{i=1}^N \delta_{X_t^{i,N}}.$$

Here:

- N is the finite approximation/population size;
- i distinguishes particles in that finite representation;
- the limiting law μ_t is unlabelled.

This has an important implementation consequence. Repeated-tree field data may use a stable identifier such as `(plot, tree_id)`, but that identifier can remain in a data registry. It only belongs inside the mathematical state if future dynamics or payoffs depend directly on identity. Observable and latent attributes remain the coordinates of the limiting state.

10. Control topology and relaxed controls

For a compact action space A , strict controls are progressively measurable A -valued processes. A useful metric is

$$\Delta(\alpha, \beta) = \mathbb{E} \int_0^T d_A(\alpha_t, \beta_t) dt.$$

If the strict action set is non-convex, a relaxed control replaces an action by a probability measure on actions, an element of $P(A)$. Because A is compact, $P(A)$ is weakly compact and convex.

Relaxation supplies compactness and convexity for existence arguments. Physical categories remain discrete whenever mixtures lack a physical interpretation.

11. Implementation interpretation

An implementation should preserve the following structure:

- keep mass separate from normalized composition;
- keep discrete biological labels discrete;
- preserve the variables needed for processor compatibility;

- keep observational confidence outside physical state unless it is itself a controlled physical variable;
- maintain stable tree IDs in a data layer when repeated measurements require them.

The next chapter places stochastic dynamics on these spaces and constructs the map from living trees to harvested logs.

3 From living trees to commercial log supply

1. Conditional McKean-Vlasov growth

Fix a biological mark class such as site, genetics, and management condition. For a representative surviving tree, let

$$\Xi_t = (s_{d,t}, s_{h,t}, q_t),$$

where $s_d = \log d$, $s_h = \log h$, and q_t contains continuous quality/form states. Age follows

$$da_t = dt.$$

The paper uses the conditional McKean-Vlasov SDE

$$d\Xi_t = b_G(\Xi_t, \hat{\mu}_t, a_t, \varsigma, a_t^G)dt + \sigma_G(\cdot)dB_t + \sigma_G^0(\cdot)dW_t^0,$$

with conditional law

$$\hat{\mu}_t = \mathcal{L}(\Xi_t \mid \mathcal{F}_t^0).$$

Dynamic programming for controlled McKean–Vlasov systems with common noise is developed by [Pham and Wei \[2017\]](#) and [Djete et al. \[2019\]](#). The forestry model uses their conditional-law state formulation.

W^0 is common noise and B is idiosyncratic noise. Conditional on the common noise, the population law remains random but the common environmental path is shared.

This distinction matters in simulation. If a drought shock is intended to be a site-year event, independently drawing a new drought shock for every tree would artificially remove within-site correlation.

2. What the law dependence is doing

A McKean-Vlasov model allows the drift or diffusion of a focal tree to depend on the population distribution. A generic competition term is

$$\mathcal{C}(x, \mu) = \int K_{\text{comp}}(x, x') \mu(dx').$$

This formulation contains several special cases.

Scalar density

If $K_{\text{comp}}(x, x') = c$ is constant, competition depends only on total mass or population density. This can be useful but is biologically coarse.

Size-weighted mean field

If

$$K_{\text{comp}}(x, x') = c D(x')^2,$$

then competition is proportional to a basal-area-like moment of the stand.

Local spatial competition

If tree location is available,

$$K_{\text{comp}}(x, x') = w(\|\ell - \ell'\|)g(D(x), D(x')),$$

can represent distance-decaying and size-asymmetric competition.

Density-only competition is a special case. The law-dependent formulation also supports size-weighted and local spatial kernels when data identify them.

3. Why log-size coordinates are useful

Writing diameter and height in logarithmic coordinates gives positivity after exponentiation and allows additive Gaussian perturbations without creating negative physical sizes.

If $Y_t = \log D_t$ follows

$$dY_t = \mu_Y(Y_t, \dots)dt + \sigma_Y(Y_t, \dots)dW_t,$$

then Itô's formula gives the corresponding diameter dynamics

$$dD_t = D_t \left(\mu_Y + \frac{1}{2} \sigma_Y^2 \right) dt + D_t \sigma_Y dW_t.$$

Thus even simple additive log-space noise becomes multiplicative noise in physical diameter.

This parametrization preserves positivity. Its empirical use requires evidence for a Gaussian diffusion approximation.

4. Sources of uncertainty

The biological parametrization is left open. The market layer can be driven by a deterministic increment model, a learned transition density, a jump process, or a size-structured PDE.

When diffusion is used, separate at least three uncertainty sources:

1. **process heterogeneity** - unobserved micro-site or biological variation;
2. **common environmental shock** - climate or disturbance shared across trees;
3. **measurement error** - error in observing DBH, height, or quality.

Only the first two belong in the state process. Measurement error belongs in the observation kernel.

Sparse remeasurement weakly identifies a complex state-dependent diffusion. The examples therefore use simple noise for illustration.

5. Well-posedness

Under standard Lipschitz dependence on state and conditional law in $\mathbf{W}_2$, linear growth bounds, and continuity in the control, the conditional McKean-Vlasov SDE admits a unique strong solution on a finite horizon and its conditional law remains in $\mathbf{P}_2$.

The coefficients satisfy a Lipschitz bound in state and population law:

$$\|b(x, \mu) - b(x', \mu')\| + \|\sigma(x, \mu) - \sigma(x', \mu')\| \leq L(\|x - x'\| + W_2(\mu, \mu')).$$

This proposition establishes well-posedness for a specified growth law. Growth-law identification requires longitudinal data.

6. Finite-mass dynamics

The normalized conditional law records composition and omits population mass. Introduce harvest hazard $\lambda_t^H(\mathbf{x}; \mathbf{u}_t)$, mortality hazard $\lambda_t^M(\mathbf{x})$, and a recruitment/planting-rate measure B_t .

For a smooth test function ϕ , the finite measure evolves weakly as

$$d\langle \phi, \mu_t \rangle = \left[\langle L_t \phi, \mu_t \rangle - \langle (\lambda_t^H + \lambda_t^M) \phi, \mu_t \rangle + \langle \phi, B_t \rangle \right] dt + \text{common-noise term}.$$

This is a transport-diffusion-reaction equation. Growth moves state; mortality and harvest remove mass; recruitment adds mass.

7. Harvest as a flow into log space

Let $K_B(d\lambda \mid \mathbf{x}, \beta_t^B)$ be a bucking kernel. Conditional on a harvested Tree state $\mathbf{x}$ and bucking rule β_t^B , it returns a finite measure of logs.

Over a short decision interval, the expected log-count intensity is

$$\Lambda_t(A) = \int_{E_T} \lambda_t^H(\xi; u_t) K_B(A \mid \xi, \beta_t^B) \mu_t(d\xi).$$

This formula is the core tree-to-log map.

It says that log supply depends jointly on:

- which trees exist;
- which trees are harvested;
- how each tree is cut;
- the uncertainty encoded in the bucking/recovery kernel.

The tonne-weighted measure is

$$\tilde{\Lambda}_t(d\lambda) = m_L(\lambda) \Lambda_t(d\lambda).$$

Again, no universal volume-to-mass coefficient is required.

8. Bucking as a control problem

Let $A_B(\mathbf{x})$ be the feasible set of cutting patterns. Given a downstream log-value function $v_t^L(\lambda)$ and bucking cost c_B , define

$$J_B(\xi, \beta; v_t^L) = \int v_t^L(\lambda) K_B(d\lambda \mid \xi, \beta) - c_B(\xi, \beta).$$

An optimal pattern satisfies

$$\beta_t^{B,*}(\xi) \in \arg \max_{\beta \in A_B(\xi)} J_B(\xi, \beta; v_t^L).$$

This creates a feedback from the market layer into physical recovery. If long, high-grade logs are scarce and carry high downstream shadow value, the preferred cut can differ from the cut that maximizes volume alone.

Under measurable-graph, compactness, measurability, and upper-semicontinuity assumptions, a measurable maximizing selector exists.

9. A simple competition example

Suppose log-DBH Y_i follows

$$dY_i = \left[r(1 - e^{Y_i}/D_{\max}) - \gamma G(\mu_t) \right] dt + \sigma dB_i + \sigma_0 dW^0,$$

where

$$G(\mu_t) = \int e^{2y} \mu_t(dy)$$

is a basal-area-like moment.

This is a mean-field model: each tree responds to a competition pressure generated by the whole size distribution. A stand with the same stem density but many large competitors produces a larger $G(\mu)$.

The stylized example shows how a size-weighted competition variable enters the structural framework.

10. Implementation

The market model takes a distribution of recoverable logs, with uncertainty, as its input. Size classes, yield tables, and grade shares provide finite-dimensional approximations to that distribution.

4 Matching heterogeneous logs to heterogeneous demand

1. Processor grade sets

Let processor j have grades $r=1, \dots, n_j \wedge G$. Each grade is a measurable subset of log state space,

$$S_{jr} \subset E_L.$$

A simple grade cell can be written

$$S_{jr} = \{ \lambda : d_s \geq d_{jr}^{\min}, L_L \geq L_{jr}^{\min}, g \in G_{jr}, q_L \in Q_{jr} \}.$$

Other constraints can add upper diameter limits, taper, defect, moisture, or piece-count conditions.

If grade cells overlap, define a grade map that selects the highest-valued admissible grade. A posted per-tonne schedule is then a measurable function $p_j \wedge 0(\lambda)$ on log space.

Price is a function of the **compatible physical state** of a log.

2. Quantity-sensitive value

A processor may value the first 1,000 tonnes of a grade differently from the next 1,000. Let

$$\pi_{jr}^{\text{marg}}(m)$$

be marginal willingness to pay for an additional tonne and define cumulative value

$$\Phi_{jr}(m) = \int_0^m \pi_{jr}^{\text{marg}}(z) dz.$$

If marginal willingness to pay is nonincreasing, Φ_{jr} is concave. This preserves convexity of the continuous allocation layer.

Minimum lots, fixed activation costs, take-or-pay clauses, and increasing block prices can break that convexity and may require mixed-integer decisions.

3. Measurement error near hard thresholds

Suppose a processor requires small-end diameter at least 18 cm. With measurement error of 0.5 cm, observations of 17.9 cm and 18.1 cm provide similar evidence about the physical diameter. A deterministic threshold can therefore make small observation changes create large grade-count changes.

Let

$$a_{jr}^\eta(\lambda) \in [0, 1]$$

be the probability that log state λ is classified as grade r under a measurement-error model with scale η .

Then expected compatible supply is an integral of a bounded continuous compatibility function against the log measure. Under narrow convergence of log measures, these integrals are continuous. This is the mathematical reason for introducing the smoothed layer: it is both statistically meaningful and numerically stable.

4. Delivered surplus

For log λ , processor j , and operator k , let

$$c_t(\lambda, j, k)$$

collect non-stumpage delivered cost. The paper separates components such as felling, extraction, loading, haulage, QA, compliance, and risk.

Compatible delivered surplus is

$$s_t(\lambda, j, k) = p_j^0(\lambda) - c_t(\lambda, j, k).$$

If the processor or operator is infeasible for that log, the edge is forbidden.

This produces a sparse compatibility hypergraph containing only feasible log-processor-operator triples.

5. The partial-transport problem

Let S be available tonne-weighted log supply, D processor demand, and O operator capacity. An allocation is a nonnegative measure Γ on feasible triples.

The linear problem is

$$V(S, D, O) = \sup_{\Gamma \geq 0} \int s \, d\Gamma$$

subject to marginal inequalities

$$\Gamma_L \leq S, \quad \Gamma_J \leq D, \quad \Gamma_K \leq O.$$

The inequalities are important. The optimizer may leave low-value logs unmatched, leave some processor demand unmet, or leave operator capacity idle.

With finite supply/demand/capacity and suitable closedness and upper semicontinuity assumptions, the feasible set is convex and narrowly compact and an optimizer exists.

6. Concavity and homogeneity of matching value

For a fixed surplus kernel, the resource-value function is jointly concave:

$$V(\theta S_1 + (1 - \theta)S_2, \theta D_1 + (1 - \theta)D_2, \theta O_1 + (1 - \theta)O_2) \geq \theta V(S_1, D_1, O_1) + (1 - \theta)V(S_2, D_2, O_2).$$

Why? If Γ_1 and Γ_2 are feasible for two resource triples, their convex mixture is feasible for the convex mixture of those resources and earns the same convex mixture of values.

The linear problem is also positively homogeneous:

$$V(aS, aD, aO) = aV(S, D, O), \quad a \geq 0.$$

Scaling all resources together scales the feasible allocation and its value.

7. Dual problem and shadow values

The dual is

$$\inf_{\varphi, \psi, \omega \geq 0} \int \varphi \, dS + \sum_j \psi_j Q_j + \sum_k \omega_k K_k$$

subject to

$$\varphi(\lambda) + \psi_j + \omega_k \geq s(\lambda, j, k)$$

on every feasible triple.

At differentiable points, the optimal potentials are marginal resource values. They answer questions such as:

- What is one more tonne of this exact compatible log type worth?
- What is one more tonne of processor demand worth?
- What is one more tonne of operator capacity worth?

This creates an information-value connection. Discovering one additional tonne in a high-shadow-value region of log space can matter more than discovering many tonnes that are already abundant or incompatible.

8. Entropic/unbalanced regularization

For large discretized problems, the paper introduces an entropy-transport relaxation with KL penalties. This replaces hard equality-style transport with a smooth convex objective over finite measures and can be solved with Sinkhorn-type scaling methods.

Regularization has a cost: the solution is biased relative to the exact linear program. The numerical programme should therefore solve a subset of scenarios at decreasing regularization and compare both objective values and dual potentials.

9. Matching output as commercially useful supply

Let Γ^* be an optimal allocation. The matched flow into processor j is

$$M_j = \Gamma^*_j(\{j\}),$$

and its demand coverage ratio is

$$\chi_j = M_j/Q_j.$$

Commercially useful supply therefore depends on tree and log attributes, location, processor grades, operator capacity, and cost. Standing volume inside a radius is only a coarse upper bound.

10. Outside options and bargaining

The transport dual tells us the marginal value of resources under the allocation problem, but a negotiated seller price needs a bargaining model.

Let $o_j(\lambda)$ be the seller's best credible alternative net value and $v_j(\lambda)$ the best net value available with target buyer j . A stylized Nash split with seller bargaining weight β gives

$$p_j^{FG} = o_j + \beta(v_j - o_j),$$

when the target buyer creates positive incremental surplus.

The paper extends v_j to include grade-specific marginal willingness to pay, least delivered cost, and a search-risk premium that falls as compatible local supply becomes easier to source.

This separates four ideas that are often collapsed into one observed price:

1. processor gross value;
2. delivered operating cost;
3. alternative-buyer outside option;
4. bargaining division of surplus.

11. A caution about causal interpretation

A dual potential is a marginal resource value inside the specified allocation model. Seller payments also depend on bargaining and outside options. Repeated bids, alternative buyers, and contract variation identify cost, scarcity, and bargaining mechanisms.

12. Implementation consequence

A practical approximation can discretize log state into bins, processor grades into nodes, and operators into capacity nodes. The exact LP becomes a benchmark. As data improve, bin boundaries can be refined around grade thresholds where a small state error has large value consequences.

5 Decision making under an uncertain physical state

1. Joint state and parameter uncertainty

The physical forest map and the model coefficients are both uncertain. Plot, remote-sensing, and operational records provide partial observations of the physical state.

Let the latent physical state be

$$X_t = (\mu_t, \nu_t, \kappa_t, r_t, \zeta_t),$$

where the first three components describe tree, processor, and operator states, **r_t** is a discrete observation/coverage regime, and **zeta_t** contains finite-dimensional auxiliary states.

Let **theta** contain unknown structural parameters for growth, recovery, processor acceptance, cost, and observations.

The controller observes a history of measurements and past controls. The joint posterior is a sufficient state:

$$\pi_t(dx, d\theta) = P((X_t, \theta) \in dx \times d\theta \mid \mathcal{I}_t).$$

This posterior carries uncertainty about both **what the system is** and **how it evolves**.

2. State and observation kernels

Given action **u_t**, the paper writes

$$X_{t+1} \sim P_\theta(\cdot \mid X_t, u_t),$$

$$Y_{t+1} \sim G_\theta(\cdot \mid X_t, u_t, X_{t+1}).$$

This separation is conceptually important. The transition kernel is about the latent system. The observation kernel is about what is measured.

Examples of observations include:

- field plots;
- processor intake and grade outcomes;
- contractor productivity;
- delivered-cost components;
- repeated bids or procurement decisions;
- remote-sensing features.

A satellite-derived index, for example, should enter **G_theta** unless an empirical observation model has established how it maps into the latent tree state.

3. Belief-state sufficiency

The posterior predictive law of the next state and observation is obtained by integrating the kernels against the current posterior. After observing **Y_{t+1}**, Bayes' rule produces **pi_{t+1}**.

Thus the future conditional distribution depends on past history only through **(pi_t, u_t)**. Under the stated Borel/measurability assumptions, any admissible history-dependent policy has an equivalent policy measurable with respect to the posterior.

This is the standard POMDP reduction in a very large state space.

For the conditional-law component, viscosity formulations on Wasserstein space are available for jump diffusions, torus models, and fully second-order HJB equations [Burzoni et al., 2019, Soner and Yan, 2022, Bayraktar et al., 2025].

4. Bellman recursion on belief space

Let $L(x, \theta, u)$ be one-step loss and

$$\bar{L}(\pi, u) = \int L(x, \theta, u) \pi(dx, d\theta).$$

If $K(d\pi' | \pi, u)$ is the Bayesian filtering kernel, the finite-horizon recursion is

$$V_t(\pi) = \inf_{u \in U(\pi)} \left\{ \bar{L}(\pi, u) + \gamma \int V_{t+1}(\pi') K(d\pi' | \pi, u) \right\}.$$

Compact actions, lower-semicontinuous loss, weakly continuous belief transitions, and a lower-semicontinuous continuation value give attainment of the Bellman minimum.

The result establishes well-posedness under the stated conditions. Numerical methods must still address the dimension of the belief state.

5. The Wasserstein HJB subproblem

For a smoother biological control subproblem, suppose the state is a probability law μ on continuous tree coordinates and $V_G(t, \mu)$ is differentiable in the Lions sense.

The HJB contains:

- a time derivative;
- a first measure derivative paired with drift;
- a derivative in the state coordinate paired with local diffusion;
- a second measure derivative generated by common noise.

Schematically,

$$0 = \partial_t V_G + \inf_{a_G} \{ F_G + \langle \partial_\mu V_G, b_G \rangle + \text{diffusion term} + \text{common-noise second-order term} \}.$$

The purpose of the equation in the paper is to expose the correct infinite-dimensional derivative structure. The full belief-state market problem also contains finite-mass geometry, hard thresholds, and impulses, so a single smooth global HJB would be misleading.

6. Nested dynamic risk

Expected value alone can underweight severe downside. The paper uses a one-step conditional risk map combining expectation and CVaR:

$$\rho(Z) = (1 - \lambda_R) E[Z] + \lambda_R \text{CVaR}_{\alpha_R}(Z).$$

Inserted recursively, this creates a time-consistent nested risk criterion.

For one-period loss,

$$\text{CVaR}_\alpha(Z) = \min_z \left[z + \frac{1}{1 - \alpha} E[(Z - z)_+] \right].$$

This epigraph is convex when the underlying loss is convex in the decision.

7. Distributional robustness is a separate layer

Bayesian posterior uncertainty is conditional on the model family. A separate robustness layer addresses local structural misspecification.

The paper therefore defines an ambiguity set around the posterior predictive law using Wasserstein distance:

$$\mathcal{P}_{DR}(\pi, u) = \{Q : W_2(Q, \hat{P}(\cdot | \pi, u)) \leq \epsilon_{DR}\}.$$

A robust Bellman step replaces expected continuation value by a worst-case expectation over that set.

Mohajerin Esfahani and Kuhn [2018] gives tractable reformulations and finite-sample guarantees for Wasserstein ambiguity sets under standard regularity conditions.

Conceptually:

```
Bayesian uncertainty: which state/parameter inside the model is plausible?  
Robust uncertainty:  what if the predictive model itself is locally wrong?
```

8. Dual control and learning value

The action changes immediate economics and the distribution of future observations. This creates an exploration-exploitation tradeoff.

Examples:

- inventorying a site to reduce uncertainty in harvestable size classes;
- measuring processor intake to learn grade acceptance;
- instrumenting contractor operations to identify productivity/cost;
- offering a controlled contract to learn price response.

The paper defines an endogenous learning value by comparing continuation value when the next observation is used versus ignored.

An information-theoretic experiment criterion such as expected information gain can choose an experiment that learns parameters strongly. A decision-focused criterion such as expected value of sample information can choose a different experiment, because only uncertainty that changes the best action has direct decision value.

Modern computational methods for Bayesian experimental design are reviewed by Rainforth et al. [2024]; variational estimators are developed by Foster et al. [2019]. The exploration effect is the dual-control mechanism described by Meijer and Rantzer [2026].

9. A simple EVSI example

Suppose there are two possible processor acceptance rates, low and high, and two possible actions: commit to a large harvesting programme or stay small. An experiment that tells the two states apart can be valuable only if the preferred action differs between them.

If both parameter states imply the same best action, information gain can be positive while EVSI is zero. The operational criterion therefore values the effect of learning on the optimal action.

10. Regime switching as impulse control

Let $R=\{r_1, r_2, r_3\}$ represent ordered observation/coverage regimes. Moving from one regime to another incurs impulse cost $K_{\{r, r+\}}$ and maps the current posterior into the enlarged regime through an embedding/prediction operator.

The quasi-variational recursion compares three choices:

1. continue within the current regime;
2. pay the impulse cost and switch;
3. stop/abandon.

The controller switches only when the value of the enlarged feasible state/action region exceeds the value of waiting by enough to justify the cost.

This is the mathematical form of a staged evidence strategy: collect enough information before committing to an expensive expansion.

11. Computational reality

The exact posterior lives on a product of measure spaces and parameters. A production controller will almost certainly need approximation: posterior particles, parametric filters, low-dimensional embeddings, rollout, scenario trees, or model-predictive control.

The structural theory is still useful because it tells us which summaries have decision meaning. Candidate reduced belief coordinates include:

- local compatible supply by grade;
- matched-demand fractions;
- posterior moments of growth/quality;
- dual matching values;
- uncertainty around high-value specification thresholds.

The next chapters formalize the reduced-order projection and the data required to identify these objects.

6 Projection into commercial coordinates

1. Why a reduced model is necessary

The structural state contains distributions of trees, logs, processors, and operators, together with a posterior over unknown parameters. Current regional data identify only a subset of these objects.

Price, scale, cost, and break-even analysis uses a finite-dimensional **projection** of the structural state.

The reduced model should be read as

$$Z_t = \Pi_{\text{red}}(\mu_t, \nu_t, \kappa_t, \pi_t),$$

where Π_{red} keeps only the coordinates needed for today's commercial questions.

2. Asset-side value envelope

For asset account i , the paper defines a value envelope

$$V_{A,i} = A_i^{ha} \left(S_i \delta_i^{op} + V_i^{asset} \lambda_i^{loss} \right).$$

Interpretation:

- A_i^{ha} is managed area;
- S_i is annual forestry spend per hectare;
- δ_i^{op} is an information-enabled proportional operating improvement;
- V_i^{asset} is standing asset value per hectare;

- λ_{loss} is avoidable annual loss/risk.

Information creates value through operational efficiency and avoided loss. The revenue model applies a rate to managed spend and captures part of this envelope.

3. Asset monetisation and adoption

Let C_i^A be an adoption indicator and define the adopted managed-spend base

$$B_A = \sum_i C_i^A A_i^{ha} S_i.$$

Asset revenue is

$$R_A = m_A B_A.$$

For one homogeneous segment, the paper uses logistic adoption

$$a(p) = \frac{1}{1 + \exp(\eta_A p - \alpha_A)}.$$

If the monetisable base is B , revenue is

$$R(p) = B p a(p).$$

The log revenue is

$$\log R(p) = \log B + \log p - \log(1 + e^{\eta_A p - \alpha_A}).$$

Differentiating twice gives

$$\frac{d^2}{dp^2} \log R(p) = -\frac{1}{p^2} - \eta_A^2 a(p)(1 - a(p)) < 0.$$

Therefore revenue is strictly log-concave and has a unique maximizer for positive price. The first-order condition is

$$\eta_A p(1 - a(p)) = 1.$$

Solving yields

$$p^* = \frac{1 + W(e^{\alpha_A - 1})}{\eta_A},$$

where W is the principal Lambert W branch.

This result is local to the simplified one-segment model. Mixtures of customer segments, fixed packages, network effects, or dynamic learning can destroy the one-dimensional global shape.

4. Processor-side monetisation

The structural matching layer outputs matched tonnes M_j . If processor j adopts the service and pays f_j per matched/visible tonne,

$$R_S = \sum_j C_j^S f_j M_j.$$

M_j is the quantity remaining after compatibility, spatial cost, and capacity constraints are applied to regional wood supply.

Today’s simpler “commercially visible supply” assumption is therefore an approximation to an operator that has a precise theoretical meaning.

5. Cost structure

The paper uses

$$C(H, N_A, N_P, n_V, W_D, G_{act}) = F + \sum_{g \in G_{act}} F_g + a_H H^{\beta_H} + b_A N_A + b_P N_P + c_V n_V + c_D W_D + \epsilon_C.$$

Components include:

- common fixed cost F ;
- geography-specific fixed blocks F_g ;
- variable coverage cost $a_H H^{\beta_H}$;
- account-level service costs;
- verification cost;
- data/compute workload;
- residual cost shock.

This form is intentionally able to separate **fixed-cost dilution** from genuine sublinear marginal scaling.

6. Fixed-cost dilution

If

$$C(H) = F + aH^\beta,$$

then local total-cost elasticity is

$$\frac{d \log C}{d \log H} = \beta \frac{aH^\beta}{F + aH^\beta} < \beta.$$

So even if the variable component is nearly linear (**beta** near one), total cost can appear strongly sublinear at small-to-medium scale because the fixed cost is being spread over more hectares.

Three declining average-cost observations leave the variable exponent unidentified. Additional scale observations or stronger cost restrictions are required.

7. Operating leverage

Let total recurring revenue be R_H and expected recurring cost C_H . Define local elasticities

$$\beta_R(H) = \frac{d \log E[R_H]}{d \log H}, \quad \beta_C(H) = \frac{d \log E[C_H]}{d \log H}.$$

Positive operating leverage requires

$$\beta_R(H) > \beta_C(H).$$

Positive operating leverage requires revenue to grow faster than recurring cost over the relevant range.

8. Break-even frontier

At fixed scale,

$$m_A B_A + f_S B_S = C,$$

where B_S is the adopted matched-tonnage base.

For fixed bases and nonnegative rates, the set

$$\{(m_A, f_S) : m_A B_A + f_S B_S \geq C\}$$

is a convex half-space. This makes simple trade-off analysis possible: stronger asset monetisation can compensate for weaker processor monetisation and vice versa.

The convexity disappears if the bases themselves respond nonlinearly to price, network effects, or participation thresholds.

9. A risk multiplier

Let $R_H > 0$ and $C_H > 0$ be random recurring revenue and cost. For target probability τ , define the revenue multiplier

$$M_H^\tau = \inf\{k > 0 : P(kR_H - C_H \geq 0) \geq \tau\}.$$

Because revenue is positive,

$$M_H^\tau = Q_\tau(C_H/R_H),$$

where Q_τ is the left quantile.

This gives an interpretable stress metric: by what multiplicative factor would the current revenue distribution need to improve to achieve a desired chance of nonnegative recurring surplus?

10. Scope of the projection

The reduced model provides conditional projections under stated adoption, matched-supply, and cost assumptions. It supplies no causal effect of hectares on profitability. New observations replace coarse inputs with structural estimates.

7 From a structural model to an identified simulator

1. Structural parameter blocks

The paper partitions parameters as

$$\theta = (\theta_G, \theta_B, \theta_M, \theta_C),$$

where:

- `theta_G` controls growth, mortality, and quality transition;
- `theta_B` controls harvest, taper, bucking, and tree-to-log recovery;
- `theta_M` controls processor acceptance, matching, price response, and bargaining;
- `theta_C` controls service, verification, and operating cost.

This decomposition is useful because each block requires different evidence. A model can be strongly identified in one block and weakly identified in another.

2. Observation model with discrepancy

The paper writes the generic observation equation as

$$Y = F_{obs}(X, u, \theta) + \delta_{model} + \epsilon.$$

Here:

- `F_obs` is the structural observation map;
- `delta_model` is model discrepancy;
- `epsilon` is measurement noise.

The distinction matters. Suppose measured DBH growth differs systematically from model predictions at hot, dry sites. If discrepancy is omitted, a Bayesian fit may force growth parameters into values that compensate for a missing mechanism. The posterior can then become precise around the wrong structural story.

Model discrepancy requires regularization and experimental designs that separate parameter effects from structural error. These restrictions preserve falsifiability.

This treatment follows the calibration framework of [Kennedy and O’Hagan \[2001\]](#).

3. Data-to-parameter identification map

The working paper proposes the following hierarchy.

Tree and plot remeasurement

Repeated DBH, height, survival, and quality/form observations inform θ_G . Repeated individual-tree measurements are especially useful because increments can be conditioned on prior state. With only two time points, process diffusion, measurement noise, and latent heterogeneity can be hard to separate.

Harvest and bucking records

Tree-level pre-harvest measurements linked to recovered log dimensions and grade outcomes inform θ_B . In their absence, the mature-tree log-recovery kernel remains uncertain because standing volume supplies insufficient information.

Contractor records

Time, equipment, fuel, payload, distance, labour, and realized productivity inform the delivered-cost kernel and operator state.

Processor intake and grading

Observed log dimensions, accepted/rejected status, realized grade, moisture, quality, and price inform processor grade maps and measurement-error-aware acceptance.

Repeated bids and alternative buyers

These help separate technical delivered surplus from bargaining and outside options.

Product conversion and renewal

Observed customer adoption, retention, and response to price inform the reduced-order demand model.

4. Identifiability is a property of the experiment

Even a mathematically identifiable parameter can be practically unidentified under a weak data design. If all observed stands have nearly the same density, for example, a competition coefficient may be confounded with site productivity. If all logs are far from a grade boundary, measurement-error scale in the grade classifier may have little support.

The control framework turns this into an experimental-design problem. A useful experiment is one that creates variation where competing parameter explanations make different predictions.

5. Particle approximation of the biological law

For a site/mark class, simulate particles

$$X_t^{1,N}, \dots, X_t^{N,N}$$

using the same common-noise path and independent idiosyncratic noise. The empirical law approximates the conditional distribution.

A practical implementation should preserve:

- shared common shock within a site-year realization;
- individual heterogeneity;
- mortality and harvest as mass-changing events;
- weights if particles represent unequal numbers of real trees.

The finite particle system is also the natural place to retain tree labels for synthetic trajectory diagnostics. The limiting measure remains unlabelled.

6. Discretizing log space

After harvest and bucking, the model produces a measure over end diameters, length, quality, moisture, biological class, and location.

A uniform grid can waste resolution far from processor thresholds and be too coarse near them. Adaptive binning is preferable where value is discontinuous or steep.

One useful hierarchy is:

```
individual tree particles
-> stochastic bucking/recovery
-> adaptive log bins
-> processor-grade nodes
-> operator-capacity nodes
```

Binning only by tonnes is insufficient if processor value changes sharply with log dimensions.

7. Exact LP as a matching benchmark

The finite discretization of the allocation problem is a linear program. Solve this exact LP on representative scenarios to establish a reference value and reference dual potentials.

For larger problems, entropy-regularized transport can trade accuracy for speed. The approximation should be diagnosed by decreasing the regularization parameter and comparing:

- primal objective value;
- matched grade quantities;
- dual potentials;
- sensitive edges near compatibility thresholds.

The regularized solver must preserve the scarcity signals used by the outer control problem.

8. Dual values as a sensitivity oracle

Suppose an outer decision changes available supply, demand, or capacity approximately affinely. Because the inner matching value is concave in these resources, optimal dual variables act as supergradients.

This enables decomposition methods. The outer problem can propose a sensing or coverage decision; the inner LP returns both value and marginal-value information; those dual values can generate cuts or local approximations for the outer search.

9. Posterior approximation

The exact joint posterior over physical measures and parameters is intractable for a large system. Candidate approximations include:

- sequential Monte Carlo over parameter and state particles;
- scenario trees;
- stochastic model-predictive control;
- rollout from a low-dimensional belief embedding;
- fitted value functions on decision-relevant summaries.

The reduction should preserve variables that affect decisions. Useful candidates from the structural model include:

- local compatible supply by grade and radius;
- matched-demand fractions;
- posterior moments of size and quality;
- high-value dual potentials;
- uncertainty near specification boundaries.

Raw hectares remain a useful exposure variable. Grade-resolved supply, matched demand, and posterior uncertainty provide additional belief coordinates.

10. A nested simulation path

For posterior draw $\mathbf{s}$:

1. sample structural parameters and a common environmental path;
2. propagate tree particles;
3. simulate mortality and harvest;
4. optimize or sample bucking;
5. convert recovered logs to mass using state-dependent density/volume;
6. construct feasible processor/operator edges;
7. solve matching and retain primal/dual outputs;
8. simulate observations generated by the chosen action;
9. update the posterior;
10. evaluate risk and continuation value.

This procedure defines a stochastic program over posterior and process distributions.

Scenario-count diagnostics can use sample-average approximation bounds [Kleywegt et al., 2002]. Correlated input sensitivity can be summarized with Shapley effects [Iooss and Prieur, 2019].

11. Decompose numerical uncertainty

At least five error sources should be tracked separately.

Biological discretization error

Finite particle number, time step, and state approximation.

Filter/posterior approximation error

Finite posterior particles, approximate likelihoods, or belief compression.

Recovery model error

Taper, defect, bucking, and log-quality uncertainty.

Matching/scenario error

Discretization of log space, entropy regularization, and finite scenario count.

Structural discrepancy

Real mechanisms missing from the model family.

Additional Monte Carlo samples reduce sampling error and leave structural discrepancy unchanged.

12. Minimum reproducibility standard

A numerical experiment should record:

- code revision;
- parameter source and whether synthetic, empirical, or prior;
- random seed where relevant;
- particle count and time step;
- discretization/bins;

- matching solver and tolerances;
- regularization value if used;
- posterior approximation method;
- output uncertainty by source where feasible.

This is the bridge from mathematical theory to a research-grade implementation.

8 East African prior-predictive calibration

1. Calibration status

The numerical results form a **prior-predictive calibration**.

It reports the outcome distribution implied by the current assumptions and uncertainty distributions. The regional tree, log, and matching operators still require identification and posterior-predictive validation.

2. Evidence used as scale anchors

The working paper reports approximately:

- 338,555 planted hectares in Uganda;
- 588,000 planted hectares in Tanzania;
- about 0.93 million hectares combined;
- 37 Chinese-owned processors in the internal intelligence base;
- about 5 million m³/year installed capacity;
- about 3 million m³/year current use;
- typical utilisation around 50-60%.

These figures are 2025–2026 evidence inputs cited by the paper. They retain the status and reference date of their underlying sources.

The paper also uses a Uganda CFR register with about 94,820 recorded planted hectares across 3,294 positive planted portfolios. The reported planted-area Gini is approximately 0.803. A single average estate size therefore gives a poor mapping from hectares to account count.

3. Missing structural state

The current calibration lacks a region-wide joint tree-level size-quality distribution. Regional data therefore leave the full mapping

$$\mu_t \rightarrow \tilde{\Lambda}_t \rightarrow \Gamma_t^*$$

unidentified. The reduced model uses priors for commercially available flow, processor coverage, adoption, monetisation, and correlated market state.

This should be read as a placeholder architecture with explicit destinations for future evidence.

4. Prior-predictive scale results

The paper reports 50,000 joint prior-predictive draws. The draw count controls Monte Carlo error conditional on the model. Prior validation and structural discrepancy require observed outcomes.

Key reported values are:

Metric	100k ha	500k ha	1m ha
Revenue P20, USDk/yr	49	326	650
Revenue P50, USDk/yr	78	451	850
Revenue P80, USDk/yr	115	613	1,090
Base OPEX P50, USDk/yr	219	424	669
Robust OPEX P50, USDk/yr	350	617	875
Joint operating surplus P20, USDk/yr	-197	-125	-66
Joint operating surplus P50, USDk/yr	-145	21	171
Joint operating surplus P80, USDk/yr	-97	191	436
P(operating surplus > 0)	0.01	0.55	0.72
Paying-ha penetration	0.25	0.26	0.262
Asset fee, USD/paying ha	2.16	2.17	2.18
Supply fee, USD/visible t	0.253	0.259	0.260

These outputs are conditional on the paper’s mapping from coverage to commercial activity and carry no causal interpretation for hectares.

5. Cost identification example

The three median base-cost anchors are approximately

$$(100k, 219k), \quad (500k, 424k), \quad (1m, 669k).$$

A naive pure power model

$$C(H) = aH^\beta$$

fits these points with an apparent exponent near 0.47.

But the paper gives an exact three-point fit of the form

$$C(H) = 162.8 + 56.2 \left(\frac{H}{100000} \right)^{0.955}$$

in USD thousands per year.

The variable exponent is then close to one. The low apparent total-cost elasticity comes largely from spreading the fixed USD 162.8k component.

This is an important model-audit example: a scaling curve can have very different structural interpretations that agree at the observed anchors.

6. Break-even thresholds

Holding the other revenue side fixed at its current median, the working paper reports one-lever deterministic break-even thresholds.

Scale	Cost mode	Asset monetisation (% managed spend)	Supply fee (USD/t)	Paying-ha penetration
100k	Base	2.23	2.04	90.3%
100k	Robust	3.73	3.70	150.9%
500k	Base	0.56	0.21	23.5%

Scale	Cost mode	Asset monetisation (% managed spend)	Supply fee (USD/t)	Paying-ha penetration
500k	Robust	0.98	0.54	41.3%
1m	Base	0.42	0.08	17.9%
1m	Robust	0.65	0.29	27.3%

A penetration threshold above 100% shows that the selected lever has no feasible break-even point under those assumptions.

7. Robustness multiplier

For an 80% target probability of nonnegative recurring operating surplus, the paper reports revenue multipliers of approximately

$$M_{100k}^{0.8} = 4.68, \quad M_{500k}^{0.8} = 1.37, \quad M_{1m}^{0.8} = 1.10.$$

This measures distance from the target under the current joint distributions. It supplies no causal effect of scale.

8. Adoption priors

The appendix gives size-dependent asset-adoption priors at central price/value conditions:

Portfolio size	1-10	10-50	50-150	150-500	500-1,500	1,500-5,000	5,000+
Base adoption	0.1%	0.5%	5%	15%	35%	60%	80%

These values are priors. Observed conversion and renewal data will replace them.

9. What new evidence would structurally improve the calibration?

The model identifies specific targets:

- repeated tree/plot measurements for growth, mortality, and quality transition;
- harvest-linked tree-to-log records for recovery and bucking;
- operator logs for productivity and delivered cost;
- processor intake for grade acceptance and density conversion;
- repeated bids/alternative buyers for outside options and bargaining;
- observed platform conversion and renewal for willingness to pay.

Each dataset has a specified target operator and likelihood.

10. Interpretation discipline

The empirical programme follows this sequence:

```
prior-predictive scale experiment
-> targeted data collection
-> operator identification
-> posterior predictive validation
-> decision experiments
-> stronger operational claims
```

The calibration is therefore a research instrument as much as a business-case output.

9 Research programme

1. The full structural chain

The model can be summarized as a composition of operators:

$$\mu_t \xrightarrow{\mathcal{G}} \mu_{t+1} \xrightarrow{\mathcal{H}, \mathcal{B}} \tilde{\Lambda}_t \xrightarrow{\mathcal{M}} \Gamma_t^* \xrightarrow{\mathcal{P}} \text{economic outcomes} \xrightarrow{\mathcal{O}} Y_{t+1} \xrightarrow{\mathcal{F}} \pi_{t+1}.$$

Interpretation:

- $\mathcal{G}$: stochastic growth, mortality, recruitment, and quality transition;
- $\mathcal{H}$: harvest selection;
- $\mathcal{B}$: bucking/recovery from tree to logs;
- $\mathcal{M}$: processor/operator matching;
- $\mathcal{P}$: price, surplus, and bargaining layer;
- $\mathcal{O}$: observation process;
- $\mathcal{F}$: Bayesian filtering/update.

The controller chooses actions that can influence several of these operators at once.

This operator view is the most reusable part of the theory. Each operator can be implemented first as a coarse approximation and later replaced by a richer, identified model while preserving the interfaces around it.

2. Why separation creates falsifiability

Suppose a predicted forest-gate price is wrong. A scalar model may provide no clear diagnosis. The structural model asks where the error entered:

- Was standing size/quality wrong?
- Was tree-to-log recovery wrong?
- Was processor compatibility wrong?
- Was haulage or operator productivity wrong?
- Was the outside option wrong?
- Was bargaining behavior wrong?

An accurate EO-derived biomass estimate can coexist with a poor commercial-supply prediction when a downstream operator is misspecified.

This decomposition turns model error into research questions.

3. Mathematical open problems

3.1 Joint finite-mass dynamics with common noise

The paper combines a normalized conditional McKean-Vlasov law with a mass-changing finite measure. A deeper theory could formulate the full stochastic transport-diffusion-reaction process directly on an HK-like finite-measure space with common noise, controls, and killing/recruitment.

Questions include existence, stability, propagation of chaos for weighted particles, and a useful dynamic-programming principle on the resulting space.

3.2 Stratified geometry

Species, genetic classes, and regimes are discrete; size, location, and quality are continuous. The global state is a stratified space.

An open problem is to characterize optimization and viscosity solutions on this hybrid geometry without pretending that discrete class changes are continuous transport.

3.3 Measure-valued impulse control

Harvest and regime changes can be impulsive. A rigorous quasi-variational inequality on a belief space whose physical component contains finite measures would connect the paper’s formal recursion to a stronger analytical theory.

3.4 Endogenous bucking and matching

When the bucking value function is taken from downstream matching duals, physical recovery and market allocation become coupled. Conditions for fixed points, stability, or decomposable solution methods are natural theoretical questions.

4. Statistical open problems

4.1 Growth diffusion versus heterogeneity

Repeated-tree data are needed to distinguish process noise, measurement error, site effects, genetic effects, and latent micro-site variation. Cross-sectional spread alone supplies no estimate of process diffusion.

4.2 Competition identification

The structural model allows a general law-dependent interaction. Which summary or kernel is identifiable from East African trial and plantation data is an empirical question.

Candidates include:

- stem density;
- basal area;
- stand-density index;
- size-asymmetric crowding;
- spatial neighbour kernels.

Out-of-site predictive performance and biological interpretability determine the appropriate level of complexity.

4.3 Recovery and grade transition

The tree-to-log kernel is a major missing link. It requires pre-harvest tree state joined to log dimensions, defect, moisture/density, grade, and processor acceptance.

4.4 Observation discrepancy

EO, field plots, management records, and processor intake observe different functions of the latent state. Cross-sensor reconciliation and structural model-discrepancy modelling are open statistical tasks.

5. Economic and market open problems

5.1 Processor demand response

Posted prices and marginal willingness to pay can differ. Quantity response, minimum lots, stockout risk, and production schedules can make demand state-dependent and non-convex.

5.2 Outside options and bargaining

The stylized Nash split is a placeholder until repeated transactions, bids, and alternative-buyer observations identify a stable bargaining structure.

5.3 Network effects and adoption

A two-sided forestry intelligence platform may become more valuable as more asset owners, processors, and operators participate. That can destroy the simple one-segment log-concavity result and create multiple equilibria or threshold behavior.

6. Decision-science open problems

6.1 Which measurements are worth buying?

The posterior framework allows a direct question: where does one additional field plot, processor intake measurement, or contractor observation have the highest expected decision value?

The matching dual suggests a powerful heuristic: prioritize uncertainty in regions of state space with high marginal supply value and high classification uncertainty.

6.2 When should coverage scale?

Regime switching depends on posterior evidence and the irreversibility of scale costs. The practical research task is to define observable sufficient statistics for a tractable control rule.

6.3 How much model complexity pays for itself?

Model selection maximizes expected decision improvement net of sensing, compute, and operating cost. This criterion can favor a simpler model with reliable input data.

7. A proposed research sequence

A sensible sequence implied by the paper is:

1. establish stable canonical state and provenance contracts;
2. assemble repeated-tree and harvest-linked recovery data;
3. identify growth, mortality, and recovery models with explicit uncertainty;
4. build processor acceptance and delivered-cost observations;
5. benchmark the discrete matching LP and its duals;
6. connect high-value uncertainty to targeted field verification;
7. fit reduced belief coordinates for sequential decisions;
8. test regime-switching rules in retrospective or simulated experiments;
9. evaluate Bayes-adaptive operational performance against a fixed benchmark.

8. A theorem/proof agenda

The current working paper contains existence, convexity, continuity, and structural propositions. A more mature mathematical paper could deepen the proof programme around:

- convergence of weighted finite-particle approximations under killing and recruitment;
- stability of the tree-to-log measure map with respect to perturbations in tree state and bucking kernels;
- differentiability or subdifferential structure of matching value with respect to supply measures;
- error bounds for entropy-regularized matching duals;
- posterior consistency under designed observation policies;
- conditions under which low-dimensional belief summaries are approximately sufficient for the outer control problem.

These should be treated as open until complete assumptions and proofs are written.

9. Product/research boundary

The product should be allowed to use simpler models than the theory whenever the simplification is explicit and empirically honest.

A production output might use a deterministic yield table while the research branch evaluates stochastic size distributions. It might use observed processor grade tables before a full quantity-response model exists. Scheduled verification can approximate the EVSI policy.

The key standard is traceability:

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production approximation
-> structural operator it approximates
-> evidence supporting the approximation
-> known failure modes
-> path to replacement
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This traceability connects each production approximation to the research model.

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